

Fake Discount Detector Using Web Scraping and Rule-Based Analytics

Abstract

This project aims to identify potentially misleading discounts on e-commerce products by extracting product information directly from product URLs and analyzing pricing attributes. The system automates product data collection, evaluates discount claims, and presents insights through an interactive dashboard.

Problem Statement

Online shoppers frequently encounter large discount percentages that may not accurately represent genuine savings. Consumers often lack the information needed to determine whether a discount is truly beneficial. This project addresses that challenge through automated data extraction and pricing analysis.

Objectives

- Extract product information from e-commerce product URLs.
- Analyze product pricing, MRP, and discount percentages.
- Detect suspicious discount patterns using business rules.
- Present results through a user-friendly dashboard.
- Support data-driven purchasing decisions.

Technology Stack

Python, Flask, Selenium, BeautifulSoup, HTML, CSS, JavaScript, ReportLab.

System Architecture

Product URL → Product Information Extraction → Data Cleaning → Pricing Analysis → Rule Engine → Dashboard Visualization → Final Verdict

Methodology

The application accepts a product URL, extracts the product title, current price, MRP, discount percentage, and product image. Extracted data is cleaned and normalized before being evaluated by a rule-based engine that identifies suspicious pricing behavior.

Rule-Based Detection Logic

1. Compare current price against historical or reference values.
2. Validate discount percentages.
3. Detect unusually inflated MRP values.
4. Identify overpriced products.
5. Generate a final classification such as Genuine, Overpriced, or Suspicious.

Dashboard Features

- Product information display
- Product image preview
- Discount analysis
- Pricing insights
- Interactive web interface
- Real-time result generation

Results and Benefits

The solution automates product evaluation, reduces manual verification effort, improves transparency in online shopping, and demonstrates practical applications of web scraping and data analytics.

Future Enhancements

- Machine Learning-based discount prediction
- Multi-platform product analysis
- Advanced price trend visualization
- Cloud deployment and public web hosting
- Browser extension integration

Conclusion

The Fake Discount Detector demonstrates how web scraping, data processing, and rule-based analytics can be combined to provide meaningful insights into product pricing. The project highlights practical software engineering skills, automation techniques, and analytical decision-making.